

VULNERABILITY ANALYSIS LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
Instructor:	Ray Lampano, Jnr.
Section Number:	15480/15481
Student Name:	Donaven Bruce
Student ID:	[Redacted]
Submission Platform:	Canvas
Report Date:	2026-02-02
Report Version:	1.0
Lab Title:	Vulnerability Analysis
Module Reference:	Vulnerability Analysis
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-02-02
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation). - The work centered mainly on PLABKALI01 for vulnerability scanning tools (Nikto, OpenVAS, Lynis) and PLABWIN10 for Windows-based enumeration tools (Hyena, SolarWinds IP Network Browser / MIB Viewer), with scans and enumeration targeting the lab topology. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The module focused on vulnerability scanning concepts, web application scanning, system auditing, false-positive analysis, and result mapping using Nikto, OpenVAS, Lynis, Hyena, and SNMP tools. - Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations. - Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline. - The module aligned most closely with exam objective 5.2 (Information Security Assessment Methodologies).	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No production infrastructure, third-party systems, or live services were contacted. - The actions documented here were performed for defensive understanding, vulnerability assessment, and ethical hacking education within the course environment. - Restricted material, live credentials, and personal data are omitted or redacted from this report.	
Work Objectives:	
- Review the basic components of vulnerability scanning, web application scanning, system auditing, false-positive identification, and result mapping. - Use Nikto, OpenVAS, and Lynis on PLABKALI01 to perform web and system vulnerability scans. - Use Hyena and IP Network Browser on PLABWIN10 for enumeration and SNMP-based discovery. - Analyze scan results, identify false positives, and map vulnerabilities to targets.	
Vulnerability Analysis Concepts and Defensive Framing:	
- The lab identified three core phases of vulnerability analysis: scanning, result analysis, and prioritization/mapping of findings. - Scan types reviewed included web application scanning (Nikto), full vulnerability assessment (OpenVAS), and system auditing/hardening (Lynis), plus enumeration support tools. - Tools demonstrated both CLI (Nikto, Lynis) and web-based (OpenVAS) approaches, showing how scanning reveals missing patches, misconfigurations, and exposed services. - Additional methods reviewed included false-positive identification and vulnerability mapping, reinforcing that effective vulnerability management requires both technical scanning and human analysis of results rather than raw output alone.	

TECHNICAL ANALYSIS

Kali Linux Vulnerability Scanning Tools on PLABKALI01:

- On PLABKALI01, Nikto was run with `nikto -host http://192.168.0.10` to scan for web server vulnerabilities; output was also saved to `plab.html` and opened in Firefox.
- OpenVAS was set up with `openvas-setup`, the Greenbone Security Assistant web interface was accessed, a new target (PLABDC01 / 192.168.0.1) was created, a task was started via the Task Wizard, and the completed report was reviewed for severity-rated findings.
- Lynis was installed with `apt-get install lynis` and run with `lynis audit system` to perform a full system security audit, revealing hardening suggestions, vulnerabilities, and compliance issues across binaries, services, kernel, and SSH configuration.

Windows Enumeration and SNMP Tools on PLABWIN10:

- On PLABWIN10, Hyena was installed and launched, PRACTICELABS.COM domain added, Domain Controllers expanded, and Services, Sessions, and Disk Space nodes viewed to enumerate running services, active sessions, and disk usage on PLABDC01.
- SolarWinds Toolset (IP Network Browser / MIB Viewer) was installed, SNMP Service on PLABDC01 was configured with community string "public", and the MIB Table was downloaded via MIB Viewer to enumerate system information.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM

PLABKALI01 — Vulnerability Scanning Tools:

```
nikto -host http://192.168.0.10
nikto -host http://192.168.0.10 -o plab.html
firefox plab.html
openvas-setup
lynis audit system
apt-get install lynis
```

PLABWIN10 — Enumeration / SNMP Tools:

```
Hyena (Domain: PRACTICELABS.COM)
Services / Sessions / Disk Space
SNMP Service Properties (Security tab)
Community: public
MIB Viewer: 192.168.0.1 / public
```

Primary Tools and Artifacts Observed:

- Nikto web vulnerability output and HTML report, OpenVAS task creation and severity-rated scan report, Lynis system audit results with hardening suggestions, Hyena services/sessions / disk views, SNMP MIB table data, and false-positive analysis notes.
- The most important evidence of vulnerability analysis activity in the lab was the detailed Nikto and OpenVAS findings, Lynis audit report highlighting SSH and service issues, and the mapping of enumerated data to specific targets for prioritization.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered vulnerability scanning concepts, web application scanning with Nikto, full vulnerability assessment with OpenVAS, system auditing with Lynis, Windows enumeration with Hyena and SNMP tools, plus false-positive analysis and result mapping.
- The recurring theme across the module was that vulnerability scanning provides the attacker (or defender) with a prioritized list of weaknesses before exploitation begins.

MITRE ATT&CK Mapping:

- Reconnaissance: T1595.002 Vulnerability Scanning - Nikto, OpenVAS, and Lynis directly mapped to active vulnerability scanning of hosts and web applications.
- Discovery: T1592 Gather Victim Host Information - Lynis system audit and SNMP MIB enumeration revealed detailed host configuration and software details.
- Discovery: T1087 Account Discovery / T1135 Network Share Discovery - Hyena enumeration provided user, group, service, and share information.
- Execution / Persistence context: identified vulnerabilities and misconfigurations can feed directly into exploitation or lateral movement phases.

Detection Opportunities:

- Unusual volumes of web requests (Nikto patterns), OpenVAS/Greenbone connections, or Lynis-style system probes are strong indicators of vulnerability scanning activity.
- IDS/IPS signatures for Nikto user-agents, SNMP GET requests, or repeated HTTP HEAD/GET to common vulnerable paths can trigger alerts.
- Log analysis of web server access logs, firewall denies for scanning ports, and endpoint monitoring for Lynis or OpenVAS processes can identify reconnaissance early.

Defensive Controls:

- Regularly patch systems, disable unnecessary services, and implement secure configurations (e.g., SSH PermitRootLogin No).
- Deploy Web Application Firewalls, intrusion detection systems, and network segmentation to limit scanning exposure.
- Use vulnerability management platforms to track, prioritize, and remediate findings while filtering false positives through manual validation.
- Apply least-privilege principles, account lockout policies, and continuous monitoring to reduce the attack surface revealed by scans.

Completion Verification:

- Work was completed across conceptual vulnerability analysis review, Nikto/OpenVAS/Lynis scanning, Windows enumeration, and false-positive/result-mapping tasks.
- Completion was evidenced by detailed scan reports, Lynis audit output, Hyena/SNMP enumeration results, and documented false-positive analysis.
- The actual commands, tool configurations, and scan targets used in the lab are preserved directly in this report for accurate documentation and reproducibility.

Key Findings:

- 1) Vulnerability scanning is the foundation of proactive defense and reveals missing patches, misconfigurations, and exposed services before exploitation occurs.
- 2) Both web-focused (Nikto) and system-wide (OpenVAS, Lynis) scanners provide complementary visibility into the same environment.
- 3) False positives are common and require manual validation to avoid wasted remediation effort.
- 4) Enumeration tools (Hyena, SNMP) often surface the same information that vulnerability scanners later flag as high-severity findings.
- 5) The strongest defensive takeaway was that effective vulnerability management requires both automated scanning and human analysis of results for accurate prioritization.

Issues / Deviations / Notes:

- The lab used deprecated software and a legacy environment, so exact menu appearance and behavior could vary (especially OpenVAS setup time).
- OpenVAS setup and Lynis installation required additional steps and patience in the sandbox.
- Firewall and service configuration changes were performed as instructed to enable scans.

Reflection:

- This lab tied together vulnerability scanning, system auditing, enumeration, and result analysis into a complete assessment workflow.
- The strongest takeaway was that vulnerability analysis is dangerous because it bridges discovery and exploitation, making regular scanning, false-positive filtering, and rapid remediation essential for defense.